

KUBERNETES TROUBLESHOOTING RUNBOOK (PRINTABLE)

This runbook covers major K8s failure classes and the diagnostic commands with expected outputs.

1. CrashLoopBackOff

- Command: `kubectl describe pod`
- Expected: 'Last State', 'Exit Code', failing command.

2. OOMKilled

- Command: `kubectl get pod -o jsonpath="{.status.containerStatuses[*].lastState.terminated.reason}"`
- Expected: OOMKilled.

3. ImagePullBackOff

- Command: `kubectl describe pod | grep -i "image"`
- Expected: authentication or tag errors.

4. NodeNotReady

- Command: `kubectl describe node`
- Expected: NetworkUnavailable, KubeletNotReady.

5. PVC Errors

- Command: `kubectl describe pvc`
- Expected: Failed binding, no PV available.

6. DNS Resolution Failures

- Command: `kubectl exec -it -- nslookup kubernetes.default`
- Expected: DNS server reachable.

7. CNI Errors

- Command: `kubectl describe node | grep CNI`

- Expected: CNI plugin load failures.

8. Scheduler Errors

- Command: `kubectl describe pod`
- Expected: No nodes available, taints, affinity.

9. API Rate Limits

- Command: `kubectl get --raw /metrics | grep apiserver_request_total`
- Expected: High request rate.

10. ETCD Timeouts

- Command: `kubectl logs -n kube-system etcd-`
- Expected: connection timeouts.

11. Network Policy Issues

- Command: `kubectl describe netpol`
- Expected: deny rules blocking traffic.

12. Pod Eviction

- Command: `kubectl get events | grep Evicted`
- Expected: node pressure conditions.

13. Disk Pressure

- Command: `kubectl describe node`
- Expected: `DiskPressure=True`.

14. Service Discovery Failures

- Command: `kubectl exec -- curl -v :`
- Expected: normal response.

15. ConfigMap Mount Failures

- Command: `kubectl describe pod`

- Expected: volume mount errors.

16. mTLS / Secret Issues

- Command: `kubectl describe secret`

- Expected: missing or invalid certificate.

17. High Pod Restarts

- Command: `kubectl get pod -o wide`

- Expected: `restartCount > 10`.